UV reconstruction: 41-coordinate FRG scaffold

This is an auditable reconstruction framework, not a recovered historical solver.

The supplied archives report a 41-running-coupling, dimension-`<= 6` FRG truncation but do not supply its operator list, complete beta-functional system, fixed-point vector, or `41 x 41` stability matrix. This project therefore makes the missing choices explicit and refuses to fabricate unknown loop coefficients.

The primary PDF also says its Wilsonian operator inventory is restricted rather than complete and corrects earlier dimension-counting errors. See `SOURCE_AUDIT.md`; no current registry in this folder is represented as the historical 41-operator basis.

`DECLARED_TRUNCATION_V0_1.md` is the new versioned basis used for the reconstruction. It makes the action content and validation conditions explicit before any full FRG coefficient is claimed.

`FRG_PROJECTION_PROTOCOL_V0_1.md` fixes the reproducibility contract for the Hessian, ghosts, Litim regulator, operator projections, residuals, and regulator-dependence checks needed before running the full solver.

`scalar_frg_lpa.py` implements the first concrete Hessian block: the six-real- scalar constant-background mass matrix and its d=4 Litim-regulated Wetterich potential trace. It does not include gauge, fermion, metric, or ghost traces.

What is implemented

- A declared 41-coordinate registry covering gravity, Standard Model gauge/Yukawa/Higgs, two real singlets (`Phi_c`, `E`), non-minimal curvature terms, and dimension-six placeholder operators.
- A beta-system protocol: canonical scaling is populated only where unambiguous; quantum terms must be supplied in a coefficient file or a derived projection module.
- Fixed-point residual, finite-difference stability matrix, critical-exponent calculation, and the exact gravity invariant:

  `beta_gN = (2 + eta_N) gN`; hence, any nonzero `gN*` requires `eta_N* = -2`.

Deliberate limitation

`model.py` does **not** claim to compute a physical 41-coupling FRG flow. Calling `beta()` without a complete projected quantum contribution raises `IncompleteFlowError`. That failure is the scientifically correct outcome until the Hessian/regulator projection is derived.

Run

Any Python 3.10+ interpreter with NumPy works (set `PY` to yours):

```
PY=python3
$PY -m unittest discover -s tests -v
$PY solve.py --check-reported
```

The reported-point check validates only the necessary gravitational condition. It cannot certify the reported fixed point without all 41 beta functions.

Explicit derived core

`scalar_core.py` implements the one-loop MS-bar flow for the two-real-scalar, massless quartic sector in the stated potential normalization. It proves a useful negative result: that isolated real one-loop core has only the Gaussian fixed point. Any non-Gaussian scalar point must therefore come from supplied gravity, gauge, Yukawa, or higher-operator terms; this code does not assume them.

higgs_two_singlet_core.py extends this using the one-loop quartic tensor contraction to four real Higgs components plus Phi_c and E. This eliminates hand-entered portal combinatorics; it is still scalar-loop-only and cannot establish a gravity-matter fixed point.

sm_gauge_core.py implements the isolated one-loop Standard Model gauge flow under the source-stated assumption that both new scalars are gauge singlets. It correctly produces a positive hypercharge beta function and asymptotic freedom for SU(2) and SU(3); it cannot yield an interacting nonzero one-loop gauge fixed point.

perturbative_matter_baseline.py couples the derived scalar tensor flow to one-loop SM gauge flow and the dominant top-Yukawa/Higgs terms, with an RK4 integrator. It is a testable matter baseline, not an FRG gravity calculation.

canonical_scaling.py supplies the exact engineering-dimension contribution for every coordinate. Quantum projections are still required before those canonical terms form a complete beta system.

einstein_hilbert_litim_benchmark.py implements a separate, pure-gravity Einstein-Hilbert optimized-cutoff benchmark from Litim. Its coupling normalization is deliberately kept distinct from this reconstruction's gN. It includes the benchmark stability matrix and critical exponents. Run run_einstein_hilbert_benchmark.py to reproduce its checked values.

Verified derived blocks and joint fixed point

Every module below declares its regulator, gauge, and projection convention in its docstring and in DECLARED_TRUNCATION_V0_1.md nothing is guessed.

Block	Module	Convention label	Status
Einstein-Hilbert + SM matter (DEP)	gravity_matter_coupling.py	arXiv:0705.3059, type-II cutoff, table convention	Verified: FP (2 Lambda_tilde*, G_tilde*) = (-4.7975095, 1.7621271), exponents (3.961, 1.644)
Scalar LPA (mass^2, quartic, Z, xi_4)	scalar_frg_lpa.py	Litim cutoff, constant background	Verified: Gaussian fixed point; step-plateau convergence
Nonminimal curvature xi_i	scalar_tensor_nonminimal_rhs.py	NP-DE-DONDER-TYPEA, arXiv:0911.0386, GMFP background	Verified: GMFP (0.00862, 0.02375), theta = 2.143 +- 2.879i, eigenvector (0.3557 +- 0.3776i, 0.8549)
Dimension-six scalar potentials	sextic_lpa.py	same Litim trace, factor 1/2^(a+b+c)	Verified: zero-sextic limit matches quartic flow; stencil checks
R^2 sector	r2_gravity_cpr.py	arXiv:0805.2909, one-loop	Negative result recorded: beta_{a_C2} = 133/(320 pi^2) > 0 constant, no fixed point
33-coordinate joint flow + spectrum	partial_flow_41.py, joint_fixed_point_stability.py	DEP gravity + Gaussian matter + NPxi + LPA dim-6	Verified below

Joint fixed point and stability spectrum (33 derived coordinates)

The joint fixed point of the derived 33-coordinate flow is the DEP SM fixed point with Gaussian matter: (lambdaN, gN) = (-4.7975095, 1.7621271), all other derived coordinates zero. Residuals: gravity/matter exact, xi_i ~1e-16, dimension-six ~1e-11 (stencil noise floor).

Critical exponents `theta = -eigenvalue` of the quantum + canonical Jacobian (`joint_fixed_point_stability.stabili`

Classification	theta	Count	Directions
Relevant	+3.9609	1	Lambda_tilde
Relevant	+2.8554	3	mH2, mPhi2, mE2
Relevant	+2.0000	3	xiH, xiPhi, xiE
Relevant	+1.6436	1	G_tilde
Marginal	0.0000	12	g1, g2, g3, yt, yb, ytau, six quartics
Irrelevant	-0.0756	3	Z_h, Z_phi, Z_E
Irrelevant	-2.0000	10	dimension-six entries

The two gravity eigenvalues reproduce the DEP table-convention exponents (3.961, 1.644). Excluded: `a_R2`/`a_C2` (NaN at the Gaussian point; the one-loop R^2 block has no fixed point) and the six pending derivative/gauge- portal coordinates (`c_HBox`, `c_PhiBox`, `c_EBox`, `c_HF2`, `c_PhiF2`, `c_EF2`).

Verdict: within the declared truncation, the reconstructed flow has a UV-attractive joint fixed point with **8 relevant**, 12 marginal, and 13 irrelevant directions. This is a reconstruction result of the declared conventions, not a reproduction of the historical 41-point claim (8 of the 41 coordinates remain undriven and are recorded as pending).

Status and license

Theoretical reconstruction and software-validation work; no experimental result is claimed. Text, notes and data: CC BY 4.0 (see `LICENSE`); code may be reused under the same terms.